# Practice workbook · Explain how mosquitoes and houseflies spread disease

Mosquitoes carrying malaria parasites and houseflies carrying bacteria, and reasons disease levels differ between places.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-VECTOR-TRANSMISSION-DISTRIBUTION

# 1. Question 8 · part a · 3 marks

June 2019 Paper 1

Worked-example retry · Attempt part a only.

**8** *P. vivax* is one of several parasites that can cause malaria.

(a) Explain how *P. vivax* is transmitted from one person to another.

**(3)**

(b) The table shows the total number of deaths from malaria and the number of deaths caused by *P. vivax*.

It shows the data for five different regions of the world in 2015.

| **Region of the world** | Q | R | S | T | U |
|---|---|---|---|---|---|
| **Total number of deaths from malaria** | 191 000 | 800 | 3800 | 14 400 | 1200 |
| **Number of deaths caused by *P. vivax*** | 1000 | 500 | 1400 | 4900 | 700 |

(i) Compare the number of deaths caused by *P. vivax* with the total number of deaths from malaria in regions Q and R.

Include calculations to support your answer.

**(3)**

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## 1. Question 8 · part a · 3 marks · Working and answer

Working and answer

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## 2. Question 4 · part b.i · 1 mark

January 2023 Paper 1

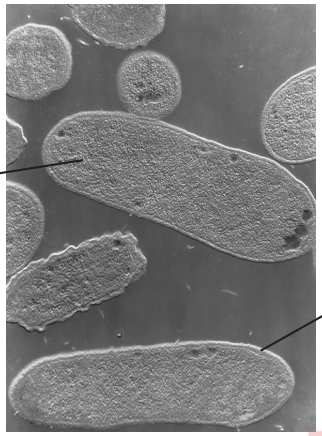
Further practice · Attempt part b.i only.

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**4** Typhoid is caused by *Salmonella typhi* bacteria.

These bacteria are spread by houseflies.

(a) The photograph shows *Salmonella typhi*, magnified 30 000 times.




(Source: © CNRI/SCIENCE PHOTO LIBRARY)

(i) Each bacterium is surrounded by cell structure Y.

Identify cell structure Y.

**(1)**

.....

(ii) On the photograph bacterium X has a length of 55 mm.

Calculate the actual length of the bacterium in micrometres (µm).

[1 mm = 1000 µm]

**(2)**

actual length = ..... µm

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## 2. Question 4 · part b.i · 1 mark

January 2023 Paper 1

Further practice · Attempt part b.i only.

(b) A team of scientists found that the legs of houseflies have hundreds of different species of bacteria on them.

(i) Flies in towns and cities carry more bacteria than flies that are found in the countryside.

Suggest a reason for this observation.

**(1)**

(ii) One species of bacterium identified is *Helicobacter pylori* which causes stomach ulcers.

Explain how houseflies could increase the transmission of *Helicobacter pylori*.

**(3)**

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## 2. Question 4 · part b.i · 1 mark · Working and answer

Working and answer

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## 3. Question 4 · part b.ii · 3 marks

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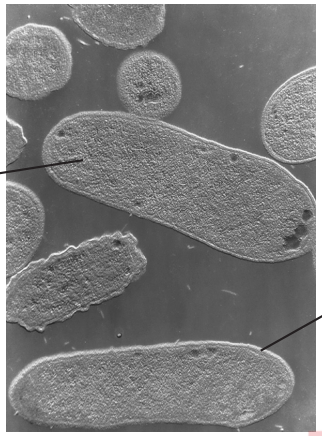
Further practice · Attempt part b.ii only.

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January 2023 Paper 1

Further practice · Attempt part b.ii only.

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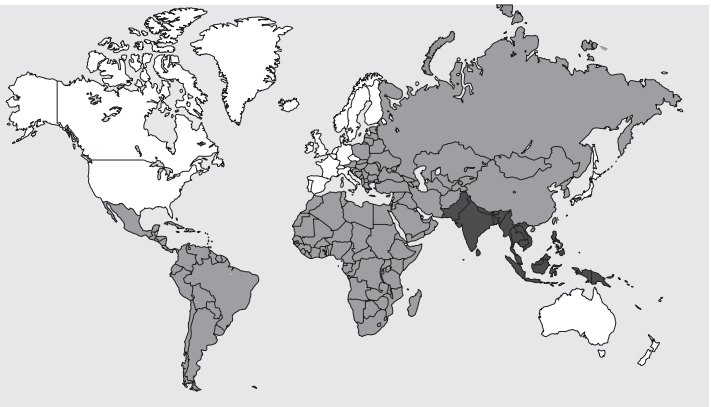
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## 3. Question 4 · part b.ii · 3 marks

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Further practice · Attempt part b.ii only.

(c) The diagram shows the levels of typhoid across the world.




(i) Give reasons for the global distribution of typhoid.

**(2)**

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## 3. Question 4 · part b.ii · 3 marks · Working and answer

Working and answer

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# 4. Question 8 · part b.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.ii only.

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(a) Explain how *P. vivax* is transmitted from one person to another.

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# 4. Question 8 · part b.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.ii only.

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(ii) Suggest one reason why the number of deaths from malaria varies across the different regions.

**(1)**

(c) Scientists are developing a vaccine to protect against malaria.

Explain how vaccinating individual people will help to protect a whole population from malaria.

**(3)**

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(d) *P. vivax* reproduces by asexual reproduction and sexual reproduction.

Explain why it is an advantage for a species to reproduce by both methods.

**(3)**

**(Total for Question 8 = 13 marks)**

## 4. Question 8 · part b.ii · 1 mark · Working and answer

Working and answer

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## 5. Question 4 · part c.i · 2 marks

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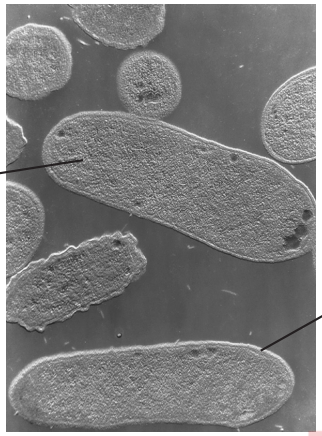
Further practice · Attempt part c.i only.

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**(2)**

actual length = ..... µm

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# 5. Question 4 · part c.i · 2 marks

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Further practice · Attempt part c.i only.

(b) A team of scientists found that the legs of houseflies have hundreds of different species of bacteria on them.

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**(1)**

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## 5. Question 4 · part c.i · 2 marks

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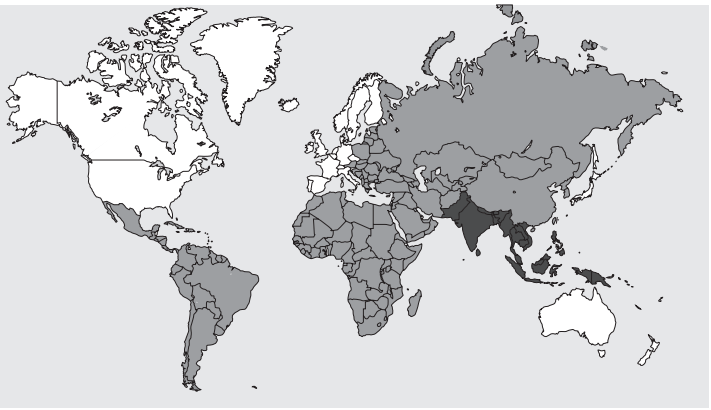
Further practice · Attempt part c.i only.

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(c) The diagram shows the levels of typhoid across the world.




(i) Give reasons for the global distribution of typhoid.

**(2)**

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## 5. Question 4 · part c.i · 2 marks

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Further practice · Attempt part c.i only.

(ii) Some strains of bacteria that cause typhoid are resistant to antibiotics.

Discuss how resistance to antibiotics develops and the effect this has on the treatment of typhoid.

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**(Total for Question 4 = 14 marks)**

## 5. Question 4 · part c.i · 2 marks · Working and answer

Working and answer

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